Phase 1 Project Proposal

International News Sentiment and Bias Analysis

DAV Semester Project

24L-2538 and 24L-2614

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1 Project Overview

The objective of this project is to design and implement an automated, end-to-end data engineering pipeline using Apache Spark through Databricks Free Edition or an equivalent Azure environment.

The project follows the Medallion Architecture, consisting of three major processing layers:

1. **Bronze Layer** – raw and traceable ingested data.
2. **Silver Layer** – cleaned, standardized, validated, and deduplicated data.
3. **Gold Layer** – curated analytical datasets and aggregations designed for business intelligence and dashboarding.

The project is designed to demonstrate how an enterprise-style data pipeline can transform semi-structured and unstructured news data into reliable analytical datasets.

2 Domain and Source Identification

2.1 Project Concept

The selected domain is **international news analytics and media monitoring**. The objective is to analyze how different international news outlets cover major geopolitical and socio-political events.

The project will examine the following major topics:

- Israel–Gaza conflict

- Russia–Ukraine war
- United States presidential elections
- Islamophobia and social conflict
- China-related geopolitical coverage

The purpose of the project is not to determine whether a particular media outlet is objectively biased. Instead, the pipeline will identify measurable differences in sentiment, topic emphasis, and language across sources. These results can then be examined by users through the final dashboard.

2.2 Data Sources

The initial dataset will contain articles and metadata collected from major international news outlets, including:

- Al Jazeera
- BBC
- CNN
- Dawn News
- Fox News
- TRT

The project will use publicly available news content and structured news feeds. The planned automated ingestion source is a combination of **news APIs, RSS feeds, and/or GDELT-based news data**, depending on availability and access limitations during implementation.

2.2.1 Primary Source Reference

The final implementation will record the exact endpoint or feed URL used for the sample data and automated ingestion.

Source references:

- GDELT Project: https://www.gdeltproject.org/
- NewsAPI: https://newsapi.org/
- RSS feeds provided by participating news organizations

The final selected source will be clearly identified in the GitHub documentation once the ingestion implementation is finalized.

2.3 Ingestion Pattern

The pipeline will support two ingestion patterns: **Full Load** and **Incremental Load**.

2.3.1 Full Load

The full-load process will be used to establish the historical baseline dataset.

The full-load process will:

1. Retrieve historical news records for the selected date range.
2. Store the original response/files in the raw data directory.
3. Add ingestion metadata such as ingestion timestamp and load type.
4. Load the records into the Bronze layer.
5. Process the data through the Silver and Gold layers.

The historical baseline is expected to contain approximately **20,000–100,000 article records**, depending on the final source availability and selected historical period.

2.3.2 Incremental Load

The incremental process will be executed periodically to retrieve newly published or updated articles.

The incremental process will:

1. Read the timestamp of the most recent successfully processed record.
2. Query the source for records published after the last successful ingestion time.
3. Store the new payload in the incremental raw-data directory.
4. Remove duplicate records using a deterministic article key.
5. Append new records and update existing records where required.

This design allows the pipeline to avoid reprocessing the complete historical dataset during every execution.

3 Data Samples and Volume Estimation

3.1 Sample Raw Data Files

The project will provide raw sample files for both required ingestion types.

The `full_load` files will represent the initial historical baseline, while the `incremental` files will contain records that were collected after the baseline period.

The incremental sample will therefore demonstrate the intended real-world workflow:

$$\textit{Historical Baseline} \rightarrow \textit{New Articles} \rightarrow \textit{Incremental Ingestion}$$

The final repository will contain the actual sample files used during development and testing.

3.2 Volume and Frequency Estimation

The expected data volume is summarized below.

Load Type	**Estimated Records**	**Estimated Size**
Full Load	20,000–100,000	50 MB–300 MB
Daily Incremental	500–5,000	2 MB–30 MB
Weekly Incremental	3,000–20,000	10 MB–100 MB

Table 1: Estimated data volume

4 Security and Compliance

4.1 PII and Sensitive Data Analysis

The raw news dataset primarily contains publicly available editorial content and metadata. It is not expected to contain highly sensitive personal information such as:

- National identification numbers
- Bank account information
- Credit card information
- Passwords
- Exact IP addresses
- Private email addresses

- Authentication credentials

However, the dataset may contain names of:

- Journalists and authors
- Public officials
- Public figures
- Individuals mentioned in news articles

These names can appear naturally within public news content. Therefore, the project will treat article content and author information as potentially sensitive public-facing information and will avoid unnecessary exposure in downstream datasets.

4.2 PII Handling Strategy

The project will follow a data-minimization approach.

The following strategies will be applied:

1. Fields that are not required for analytical processing will be removed before the Silver layer.
2. Exact article URLs will be retained in Bronze for traceability but may be excluded or masked in Gold-level reporting datasets.
3. Author information will only be retained where it provides analytical value.
4. No private identifiers or authentication credentials will be stored in the repository.
5. API keys and credentials will never be committed to GitHub.
6. Configuration files containing credentials will be excluded using `.gitignore`.

Because the project primarily uses public editorial information, the main security concern is responsible handling of public names, article content, source URLs, and API credentials rather than financial or identity-document PII.

5 High-Level Medallion Data Modeling

5.1 Architecture Overview

The proposed architecture follows the standard Medallion pattern:

Raw Source → Bronze → Silver → Gold → Dashboard

Apache Spark will be used for distributed processing and transformation.

5.2 Bronze Layer

The Bronze layer will contain the raw ingested records with minimal transformation.

Its primary objectives are:

- Preserve source data.
- Maintain data lineage.
- Support reprocessing.
- Record ingestion metadata.
- Distinguish full and incremental loads.

The Bronze layer will preserve the original source representation as much as possible.

5.3 Bronze-to-Silver Transformations

The Silver layer will convert raw records into a consistent analytical dataset.

The major transformations will include:

1. Standardizing column names.
2. Converting timestamp fields into a consistent datetime format.
3. Trimming leading and trailing whitespace.
4. Handling null and missing values.
5. Removing malformed records.
6. Removing duplicate articles.
7. Standardizing source names.
8. Standardizing topic and region labels.
9. Generating a deterministic `article_id`.
10. Normalizing article text.
11. Validating publication dates.
12. Adding data-quality indicators where required.

Duplicate articles will be detected using the source URL where available. Where URLs are missing or unreliable, a hash generated from fields such as source, headline, and article content will be used.

5.4 Gold Layer

The Gold layer will contain business-ready datasets optimized for reporting and dashboarding.

6 Business Intelligence and Dashboard

6.1 Dashboard Objective

The final dashboard will be developed using Power BI, Tableau, or Streamlit, depending on the final implementation environment.

The dashboard will consume Gold-layer data rather than directly querying raw data. This separation ensures that the visualization layer uses curated and validated datasets.

6.2 Business Questions

The dashboard will address the following questions:

1. How does article volume vary across news sources and topics?
2. How does measured sentiment vary by source and topic?
3. How does coverage of selected topics change over time?
4. Which regions receive the greatest amount of coverage?
5. Which keywords occur most frequently within different sentiment categories?

The term "bias" in the dashboard will refer to measurable differences in source-level sentiment, topic emphasis, or keyword usage. The dashboard will not claim to establish objective or intentional bias solely from sentiment scores.

6.3 Planned Visualizations

6.3.1 1. Sentiment Distribution by Source

A stacked bar chart will display the distribution of positive, neutral, and negative articles across the selected news sources.

This visualization will allow users to compare sentiment distributions without requiring a single overall ranking of news organizations.

6.3.2 2. Topic Coverage Over Time

A line chart will show the number of articles published for each major topic over time.

This will help identify changes in media attention and periods of increased coverage.

6.3.3 3. Region–Topic Heatmap

A heatmap will display article volume across regions and topics.

Darker cells will indicate higher article counts, allowing users to identify areas receiving greater coverage.

6.3.4 Additional Metrics

The dashboard may also include:

- Total article count
- Number of active sources
- Average sentiment score
- Daily article volume
- Most frequent keywords

7 Engineering Setup and FinOps

7.1 Version Control

The project will be maintained using Git and GitHub.

The repository will contain:

- Pipeline source code
- Sample raw data
- Bronze/Silver/Gold transformation code
- Documentation
- Configuration templates
- Dashboard resources
- Requirements file
- Project proposal

7.2 GitHub Repository

The final repository link will be:

https://github.com/rafayrizwanbabri/currency-exchange.git

The repository will contain the required sample files for both full-load and incremental-load ingestion.

API credentials, passwords, access tokens, and other secrets will not be committed to the repository.

7.3 Infrastructure

The project will use Apache Spark through one of the following student-access environments:

- Databricks Free Edition
- Azure for Students, where available

The final implementation environment will be selected based on availability, resource limits, and the requirements of the course.

7.4 FinOps Strategy

Because the project is intended for a student/free-tier environment, resource usage will be actively controlled.

The following practices will be applied:

1. Use small datasets during pipeline development and debugging.
2. Validate transformations on a representative sample before processing the complete historical dataset.
3. Avoid unnecessary reprocessing of historical data.
4. Use incremental ingestion for newly available records.
5. Apply filtering and partitioning where appropriate.
6. Stop or terminate compute resources when processing is complete.
7. Store only required datasets in persistent cloud storage.
8. Monitor storage and compute consumption throughout development.

The full historical dataset will only be processed after the pipeline has successfully passed testing on smaller samples.

8 Conclusion

This project proposes an end-to-end data engineering solution for international news analytics using Apache Spark and the Medallion Architecture.

The pipeline will transform raw news data into progressively refined Bronze, Silver, and Gold datasets while supporting both historical full loads and periodic incremental loads. The Silver layer will provide a standardized analytical representation of news articles, while the Gold layer will provide aggregated datasets for business intelligence.

The final dashboard will allow users to explore article volume, topic coverage, sentiment patterns, regional coverage, and keyword trends across multiple international news sources.

The project also incorporates essential engineering practices including version control, sample-data reproducibility, data minimization, credential protection, incremental processing, and FinOps-aware resource management.

Overall, the proposed solution provides a practical demonstration of how modern enterprise data engineering principles can be applied to a real-world, semi-structured data domain.